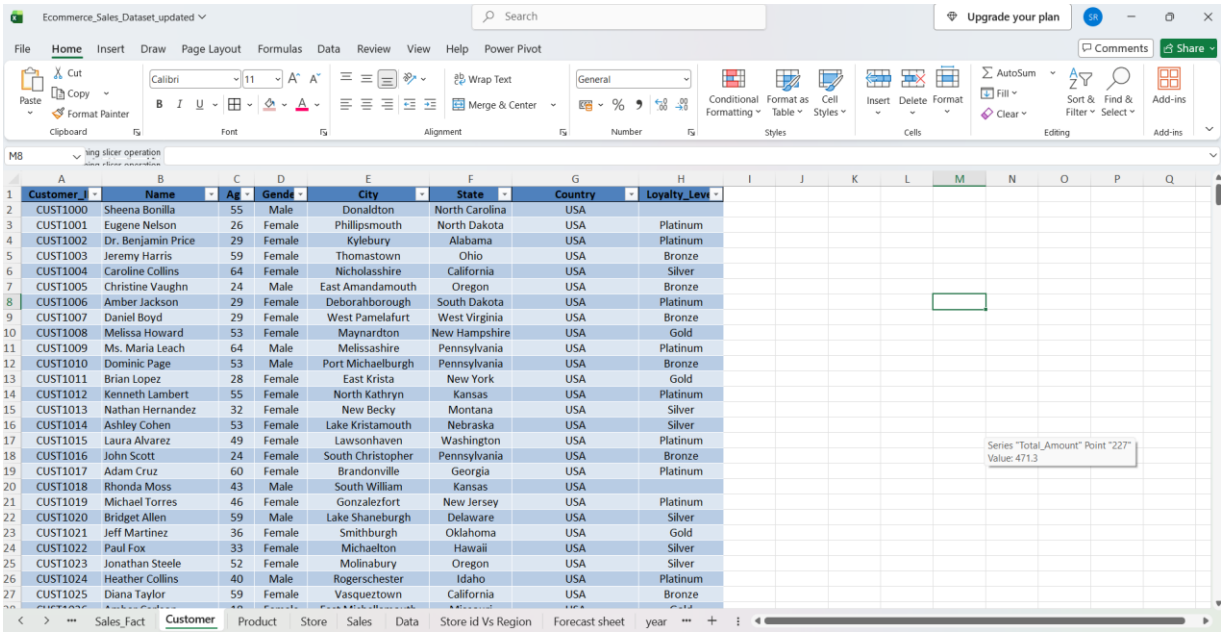
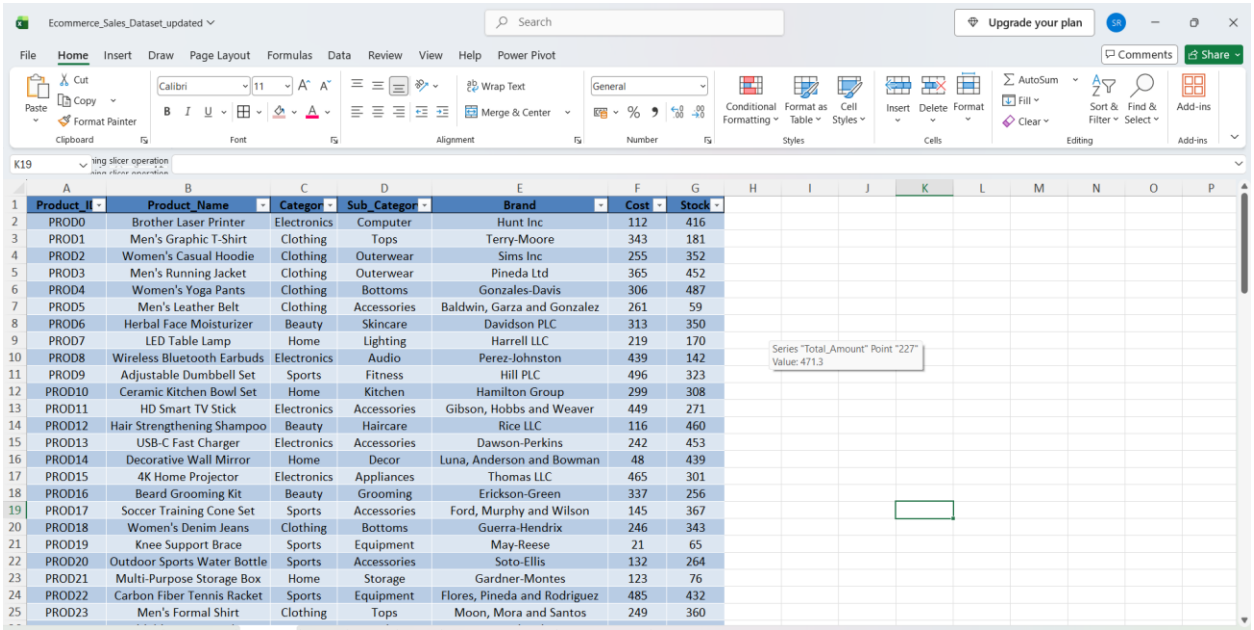
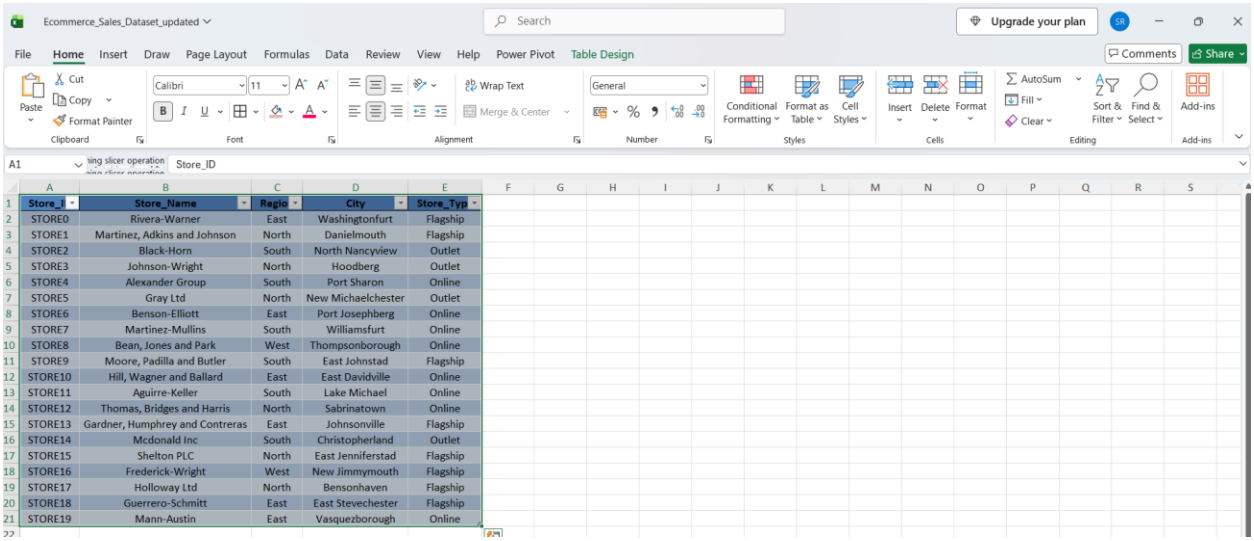
Data Entry & Organization



Customer_I	Name	Ag	Gende	City	State	Country	Loyalty_Levi
CUST1000	Sheena Bonilla	55	Male	Donaldton	North Carolina	USA	
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA	Silver
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA	Gold
CUST1022	Paul Fox	33	Female	Michaelton	Hawaii	USA	Silver
CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon	USA	Silver
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	USA	Platinum
CUST1025	Diana Taylor	59	Female	Vasqueztown	California	USA	Bronze

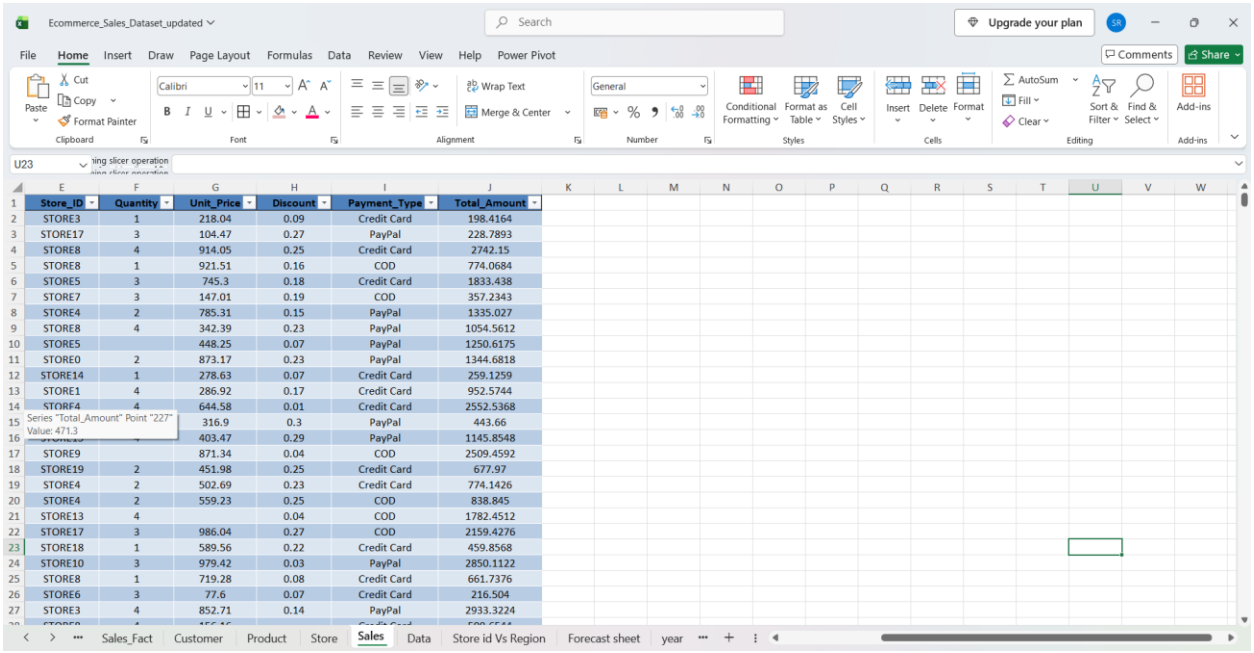


Product_II	Product_Name	Categor	Sub_Categor	Brand	Cost	Stock
PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	112	416
PROD1	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	343	181
PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc	255	352
PROD3	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	365	452
PROD4	Women's Yoga Pants	Clothing	Bottoms	Gonzales-Davis	306	487
PROD5	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzalez	261	59
PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson PLC	313	350
PROD7	LED Table Lamp	Home	Lighting	Harrell LLC	219	170
PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439	142
PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill PLC	496	323
PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	299	308
PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	449	271
PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	116	460
PROD13	USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242	453
PROD14	Decorative Wall Mirror	Home	Decor	Luna, Anderson and Bowman	48	439
PROD15	4K Home Projector	Electronics	Appliances	Thomas LLC	465	301
PROD16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337	256
PROD17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy and Wilson	145	367
PROD18	Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix	246	343
PROD19	Knee Support Brace	Sports	Equipment	May-Reese	21	65
PROD20	Outdoor Sports Water Bottle	Sports	Accessories	Soto-Ellis	132	264
PROD21	Multi-Purpose Storage Box	Home	Storage	Gardner-Montes	123	76
PROD22	Carbon Fiber Tennis Racket	Sports	Equipment	Flores, Pineda and Rodriguez	485	432
PROD23	Men's Formal Shirt	Clothing	Tops	Moon, Mora and Santos	249	360



Ecommerce_Sales_Dataset_updated

Store_ID	Store_Name	Region	City	Store_Type
STORE0	Rivera-Warner	East	Washingtonfurt	Flagship
STORE1	Martinez, Adkins and Johnson	North	Danielmouth	Flagship
STORE2	Black-Horn	South	North Nancyview	Outlet
STORE3	Johnson-Wright	North	Hoodberg	Outlet
STORE4	Alexander Group	South	Port Sharon	Online
STORE5	Gray Ltd	North	New Michaelchester	Outlet
STORE6	Benson-Elliott	East	Port Josephberg	Online
STORE7	Martinez-Mullins	South	Williamsfurt	Online
STORE8	Bean, Jones and Park	West	Thompsonborough	Online
STORE9	Moore, Padilla and Butler	South	East Johnstad	Flagship
STORE10	Hill, Wagner and Ballard	East	East Davidville	Online
STORE11	Aguirre-Keller	South	Lake Michael	Online
STORE12	Thomas, Bridges and Harris	North	Sabrinatown	Online
STORE13	Gardner, Humphrey and Contreras	East	Johnsonville	Flagship
STORE14	Mcdonald Inc	South	Christopherland	Outlet
STORE15	Shelton PLC	North	East Jenniferstad	Flagship
STORE16	Frederick-Wright	West	New Jimmymouth	Flagship
STORE17	Holloway Ltd	North	Bensonhaven	Flagship
STORE18	Guerrero-Schmitt	East	East Stevechester	Flagship
STORE19	Mann-Austin	East	Vasquezborough	Online



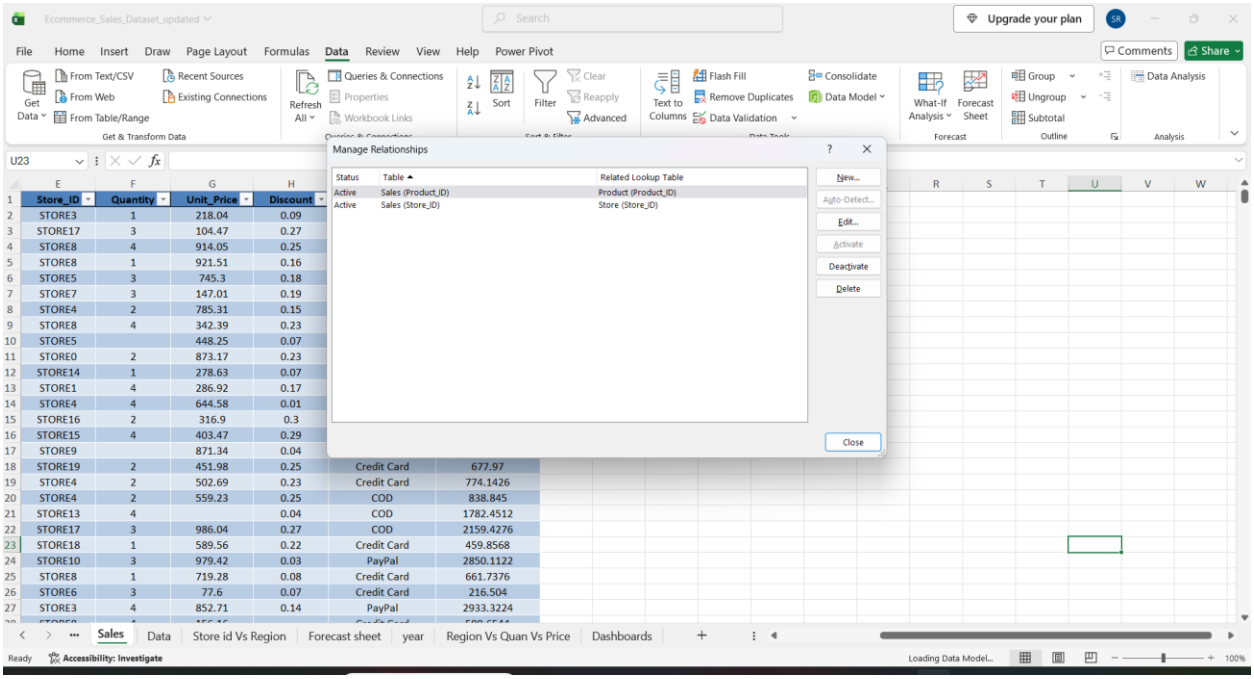
Ecommerce_Sales_Dataset_updated

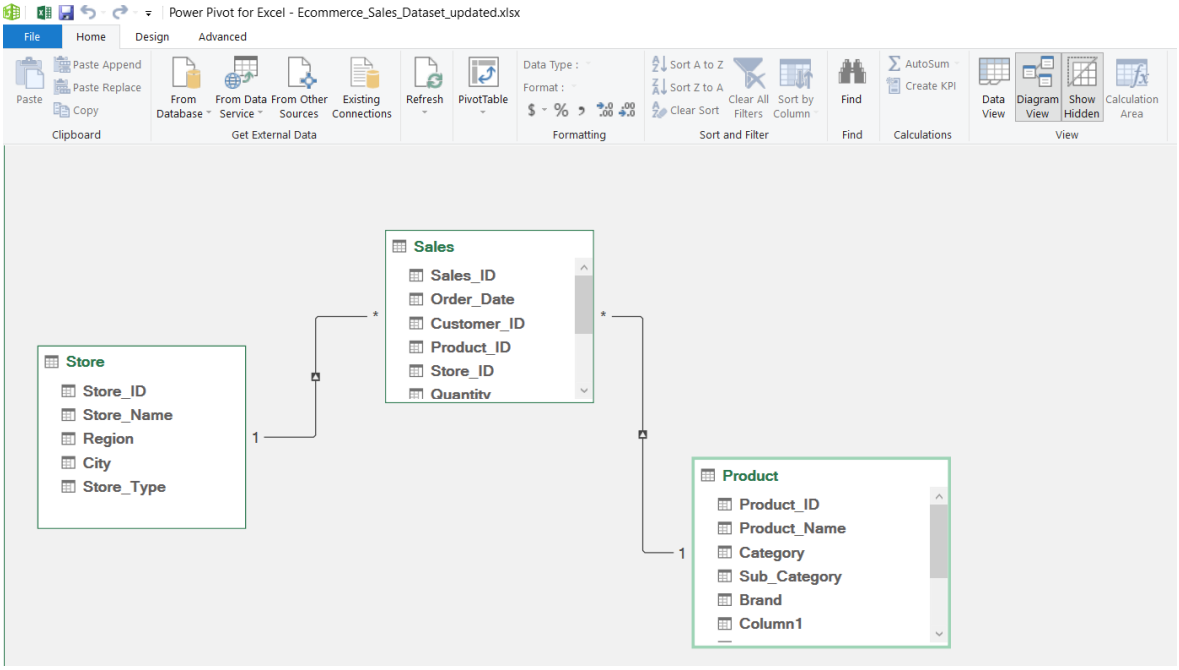
Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
STORE3	1	218.04	0.09	Credit Card	198.4164
STORE17	3	104.47	0.27	PayPal	228.7893
STORE8	4	914.05	0.25	Credit Card	2742.15
STORE8	1	921.51	0.16	COD	774.0684
STORE5	3	745.3	0.18	Credit Card	1833.438
STORE7	3	147.01	0.19	COD	357.2343
STORE4	2	785.31	0.15	PayPal	1335.027
STORE8	4	342.39	0.23	PayPal	1054.5612
STORE5		448.25	0.07	PayPal	1250.6175
STORE0	2	873.17	0.23	PayPal	1344.6818
STORE14	1	278.63	0.07	Credit Card	259.1259
STORE1	4	286.92	0.17	Credit Card	952.5744
STORE4	4	644.58	0.01	Credit Card	2552.5368
		316.9	0.3	PayPal	443.66
		403.47	0.29	PayPal	1145.8548
STORE9		871.34	0.04	COD	2509.4592
STORE19	2	451.98	0.25	Credit Card	677.97
STORE4	2	502.69	0.23	Credit Card	774.1426
STORE4	2	559.23	0.25	COD	838.845
STORE13	4		0.04	COD	1782.4512
STORE17	3	986.04	0.27	COD	2159.4276
STORE18	1	589.56	0.22	Credit Card	459.8568
STORE10	3	979.42	0.03	PayPal	2850.1122
STORE8	1	719.28	0.08	Credit Card	661.7376
STORE6	3	77.6	0.07	Credit Card	216.504
STORE3	4	852.71	0.14	PayPal	2933.3224

Series "Total_Amount" Point "227"
Value: 471.3

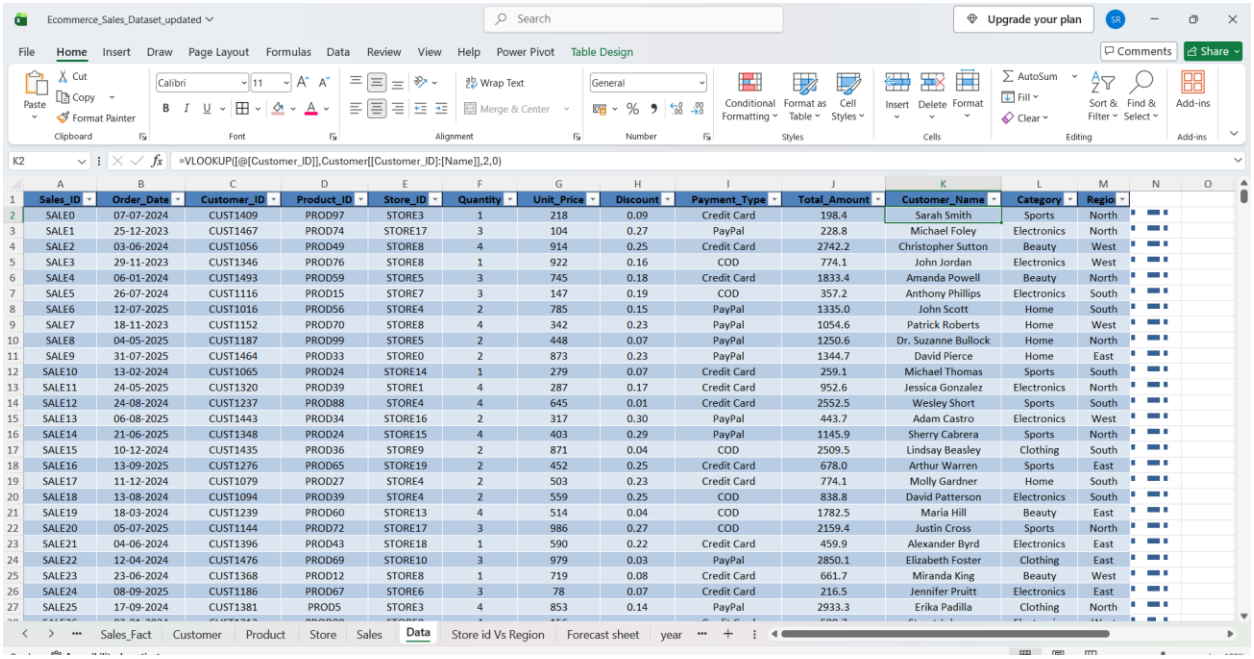
Sales_Fact | Customer | Product | Store | Sales | Data | Store id Vs Region | Forecast sheet | year

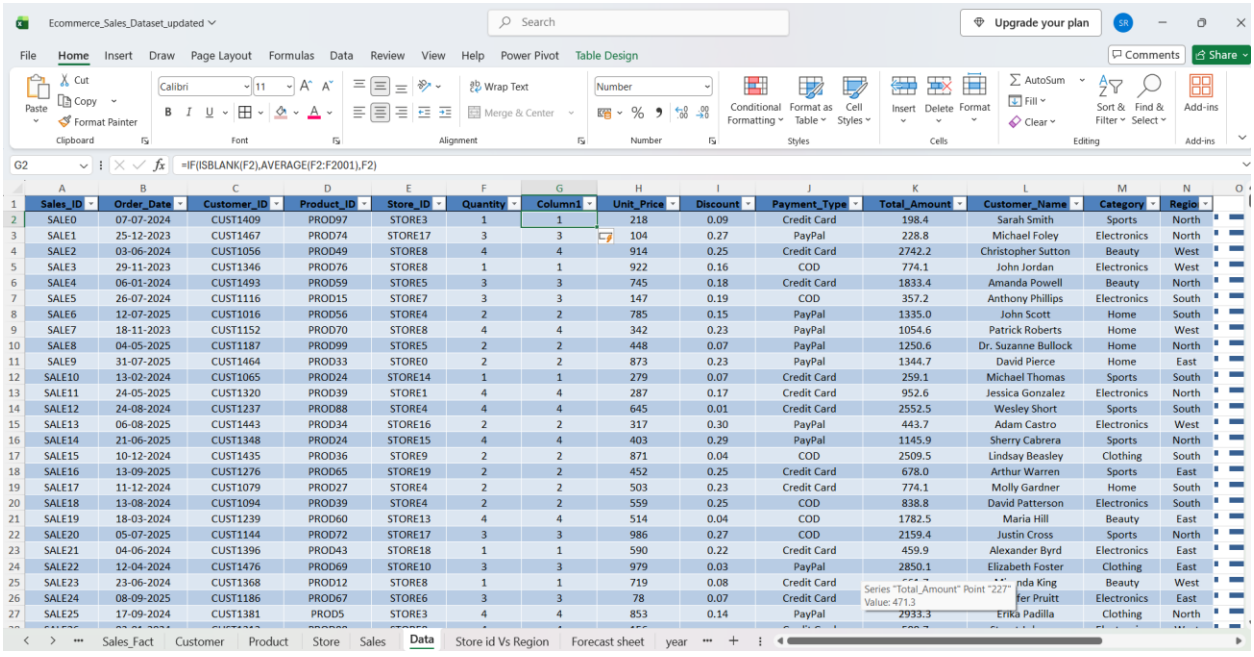
Data Cleaning & Transformation.



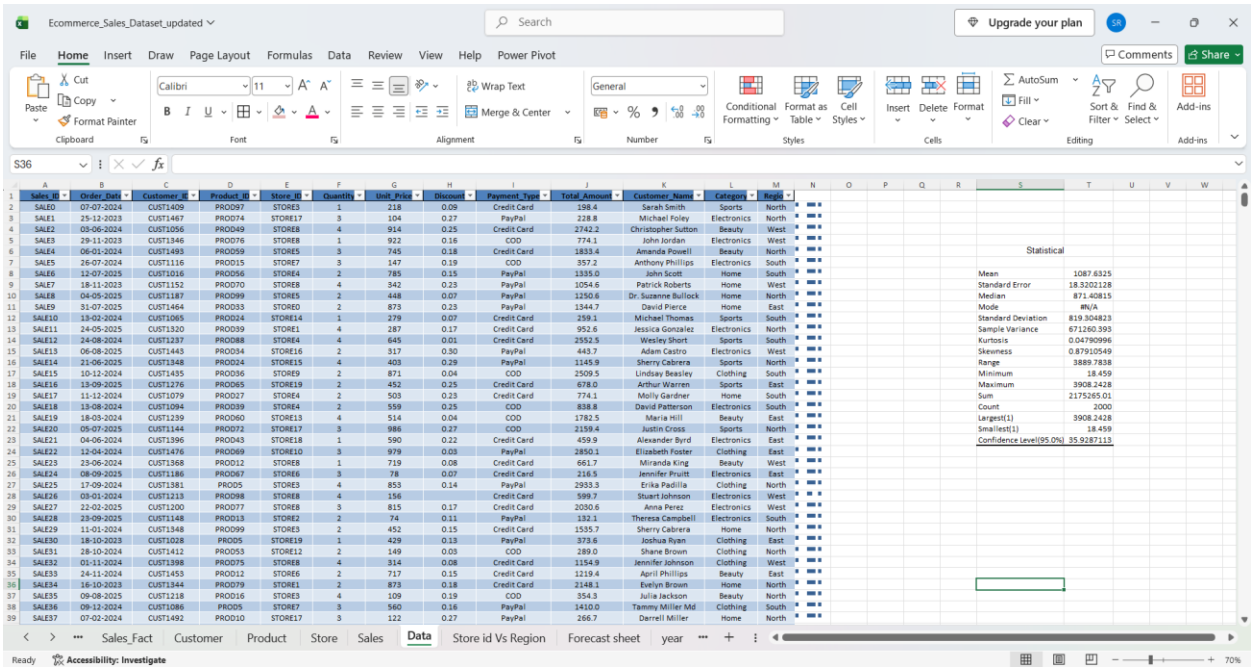


Excel Formulas & Functions



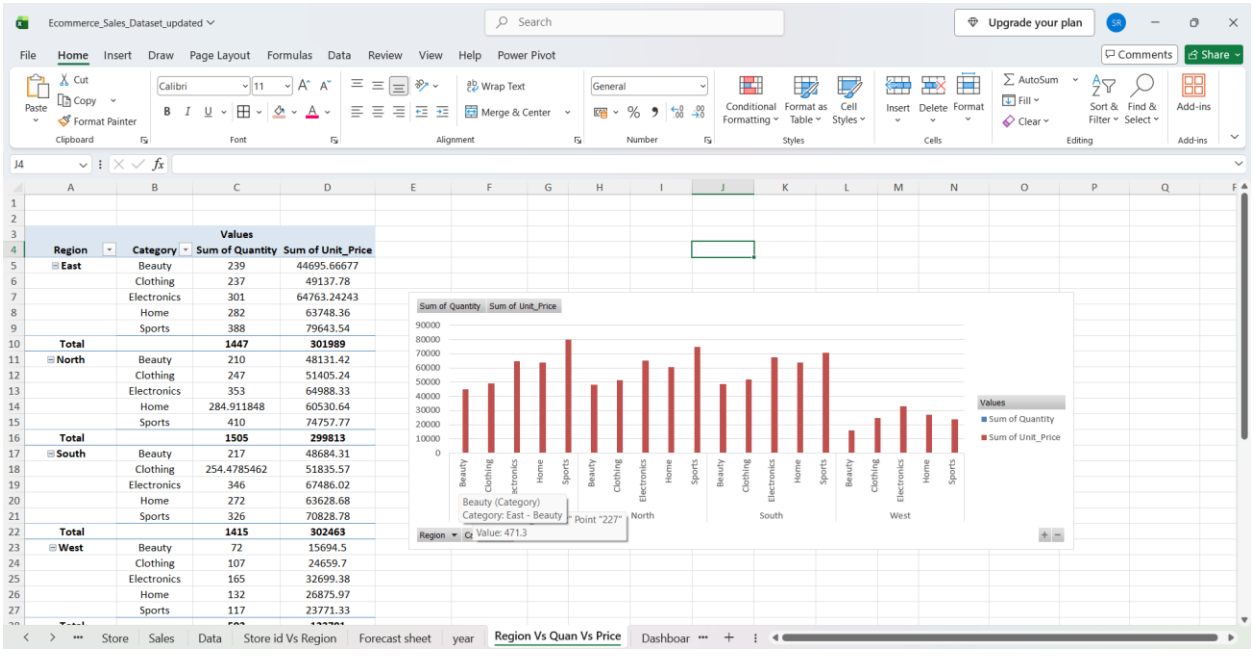


Descriptive Statistics



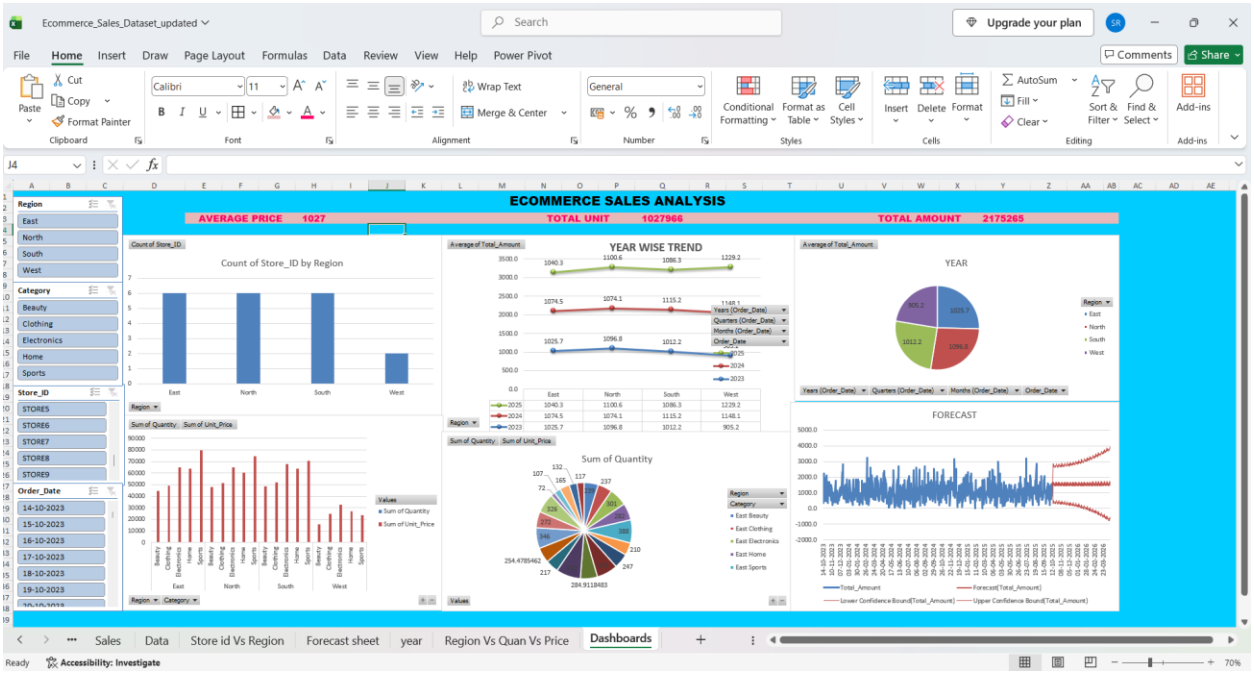
Statistical	
Mean	1087.6325
Standard Error	18.3202128
Median	871.40815
Mode	#N/A
Standard Deviation	819.304823
Sample Variance	671260.393
Kurtosis	0.04790996
Skewness	0.8791054
Range	3889.7838
Minimum	18.459
Maximum	3908.2428
Sum	2175265.01
Count	2000
Largest(1)	3908.2428
Smallest(1)	18.459
Confidence Level(95.0%)	35.9287113

Data Summarization (Pivot)



Region	Category	Sum of Quantity	Sum of Unit_Price
East	Beauty	239	44695.66677
	Clothing	237	49137.78
	Electronics	301	64763.24243
	Home	282	63748.36
	Sports	388	79643.54
Total		1447	301989
North	Beauty	210	48131.42
	Clothing	247	51405.24
	Electronics	353	64988.33
	Home	284.911848	60530.64
	Sports	410	74757.77
Total		1505	299813
South	Beauty	217	48684.31
	Clothing	254.4785462	51835.57
	Electronics	346	67486.02
	Home	272	63628.68
	Sports	326	70828.78
Total		1415	302463
West	Beauty	72	15694.5
	Clothing	107	24659.7
	Electronics	165	32699.38
	Home	132	26875.97
	Sports	117	23771.33

Data Visualization



Reporting & Insights

Overview.

The dataset contains 2,000 sales transactions from 14 October 2023 to 13 October 2025, covering 492 unique customers, 100 product IDs, and 20 stores. Total recorded sales were 2,175,266.50, with an average transaction value of 1,087.63 and 4,958 units sold.

Key Findings

• **Category performance:** Sports generated the highest sales at 541,973.50 (24.9%), followed by Electronics at 504,311.20 (23.2%). Beauty had the lowest category sales at 322,450.30 (14.8%).

• **Regional performance:** North led total sales with 647,345.30 (29.8%), closely followed by South at 635,212.10 and East at 622,032.50. West contributed 270,676.60 but had the highest average transaction value at 1,161.70.

• **Payment methods:** PayPal recorded the highest sales (756,718.90), followed by Credit Card (714,844.40) and COD (703,703.20), showing that all three methods contribute meaningfully.

• **Timing:** March 2025 was the strongest month in the dataset, with 107,589.00 in sales. October 2025 only includes data through 13 October, so it is a partial month.

Data Quality Checks

• Four Discount values are missing (SALE26, SALE52, SALE58, and SALE259); verify them from source records before discount-based analysis.

• No duplicate Sales_ID values or exact duplicate rows were found.

• A calculation check using Quantity × Unit_Price × (1 − Discount) differs from Total Amount by more than 1 for 318 of the 1,996 rows with a populated Discount. Confirm the intended business formula and investigate before changing recorded sales.

Recommended Decisions

• Prioritize stock availability and targeted promotions for Sports and Electronics, the two largest sales categories.

• Review West-region opportunities to increase transaction volume while learning from its comparatively high average transaction value; maintain strong coverage in North, South, and East.

• Keep PayPal, Credit Card, and COD available, and monitor payment-method costs and customer preferences before making changes.

• Validate missing discounts and reconcile the Total_Amount calculation rule before using the data for final financial reporting. Compare monthly performance using complete months only.

Conclusion: Sales are led by Sports and Electronics, while North contributes the largest regional total. The most important next steps are to support leading categories, investigate regional growth opportunities, and resolve data-quality questions before making financial decisions.